



5G: The Internet of the Future

The race for 10 Gbps speeds is the next big step towards the evolution of 5G. So, before the internet of the future arrives, let us delve deeper into what it entails.



mous vehicles. 5G is also supposed to be a new chapter in security and a likely nurturer of big and small scale industries, which are likely to mushroom as we move towards a utopian, connected future.

But first, a history lesson on the "G"

To understand how mobile wireless technology works, we need to dive into the genesis of the technology. We need to get into the past of the technology, we need to understand why today's technology won't be enough to sustain the outgrowing demand of data and why 5G could be the solution which may enable us to be more connected and the world to be more automated. This will also help us understand why 5G is a much bigger deal than the existing 3G and 4G networks.

1G

Let's start with 1G, the true first generation of mobile networks or wireless systems, which enabled consumers to connect with each other on a global level. Before this, there was IMTS (Improved mobile telephone service), which was severely expensive and extremely limited. A whole IMTS unit was as big as a briefcase and required a separate large antenna, plugged to a power source to place calls. This analogue technology evolved into 1G or AMPS (Advanced Mobile Phone System), which came into existence in the late 1980s. This carved the cornerstone for mobile telephony, reducing the briefcase size telephone to a handheld device. However, it had its set of problems. 1G had low capacity, poor voice links and no security. Voice calls were

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As India progresses towards a pan India 4G network rollout, the research behind the fifth generation of cellular broadband technology is rapidly moving towards completion, called 5G at its current state. 5G simply stands for 'fifth generation', and it's the upcoming standard of mobile communications which suggests that it will revolutionise how we interact with technology, moreover how technology interacts with us. So, what is 5G? Well, to be exact, it is not defined yet. It is rather, a very complex topic at the moment, since the properties and ingredi-

ents of the technology are still under consideration. However, currently, industry experts have chalked out some of its goals in the early stages of the conceptualisation of this upcoming technology. According to various organizations and research teams working on 5G, it will take at least another half a decade to become a global standard. 5G is supposed to bring lower latency, better and wider connectivity and best of all, super speeds of up to 10 Gbps. This is just the tip of the 5G iceberg as it is being conceptualised to be way bigger than the current 4G standard. It will not only help internet penetration in remote areas, but if the required goals are met, it will be an essential part of mission critical industries such as medicine and autono-